

Can the first 24 hours predict a long ICU stay?

Lung-cancer patients • eICU Collaborative Research Database v2.0

20-minute story

First-24-hour data

Fair final test



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IT Academy Final Project



Why does a long ICU stay matter?

Because the consequences are clinical, operational and human



More clinical complexity

More monitoring, procedures and risk of complications.



More pressure on ICU capacity

Beds, staff and equipment must be planned earlier.



More uncertainty for families

A clearer early estimate can support communication and planning.

The goal is not to replace clinicians — it is to create an earlier warning signal.

eICU v2.0: a national critical-care laboratory

A multicenter, deidentified database built from U.S. tele-ICU programs



200,859

ICU patient-unit encounters



139,367

unique patients



208

hospitals across the U.S.



335

intensive-care units



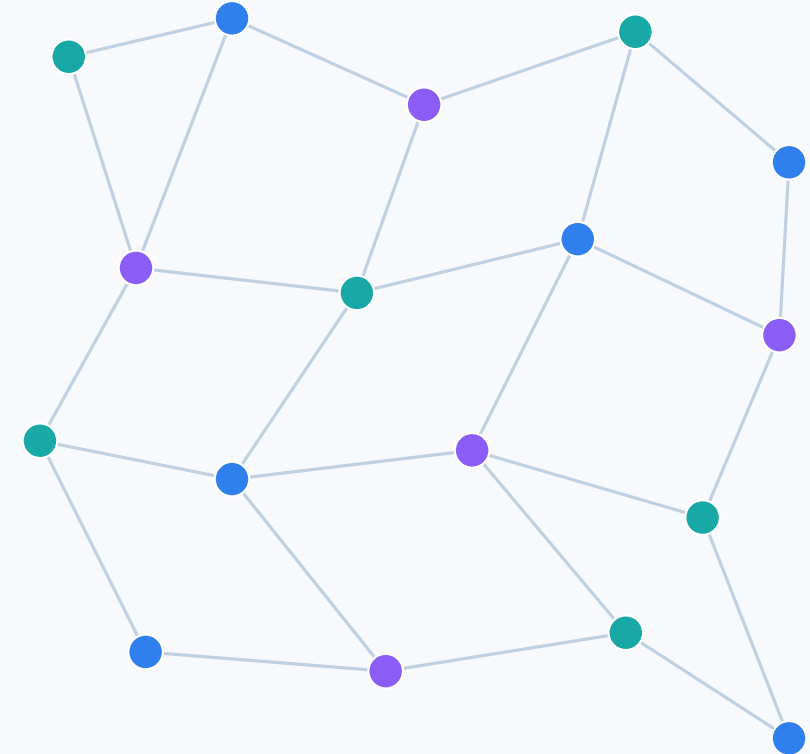
3

linked relational tables



2014–2015

admission years



One database, many hospitals

Sources: Pollard et al. (2018); PhysioNet eICU-CRD v2.0.

Why researchers trust and reuse eICU

Scale is useful only when the data are responsibly shared and well documented



accesses to the Scientific Data descriptor



citations to the foundational database paper



deidentified under Safe Harbor rules



human-subject research training required



credentialed access and data-use agreement

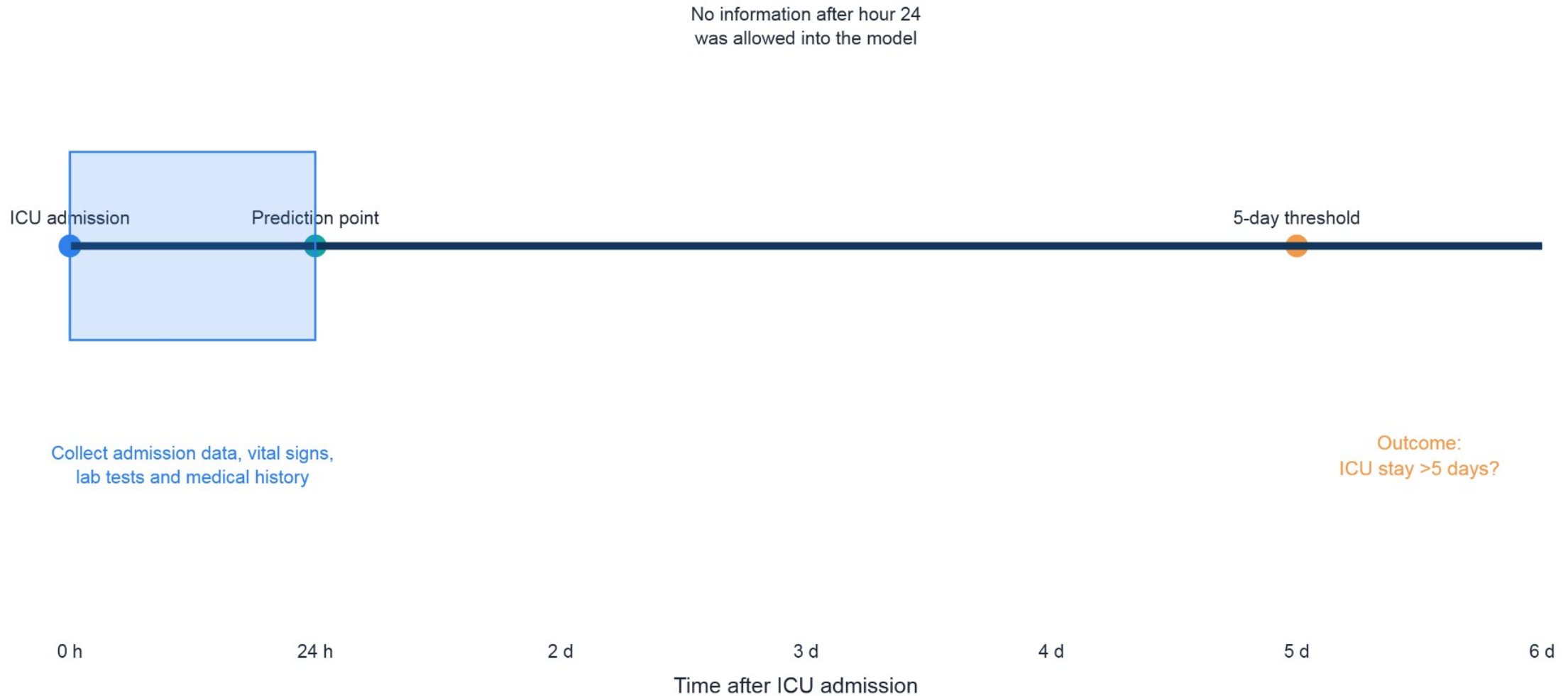


public documentation and community SQL notebooks

Citation count is a verified impact indicator — not an exact count of every paper that has used eICU.

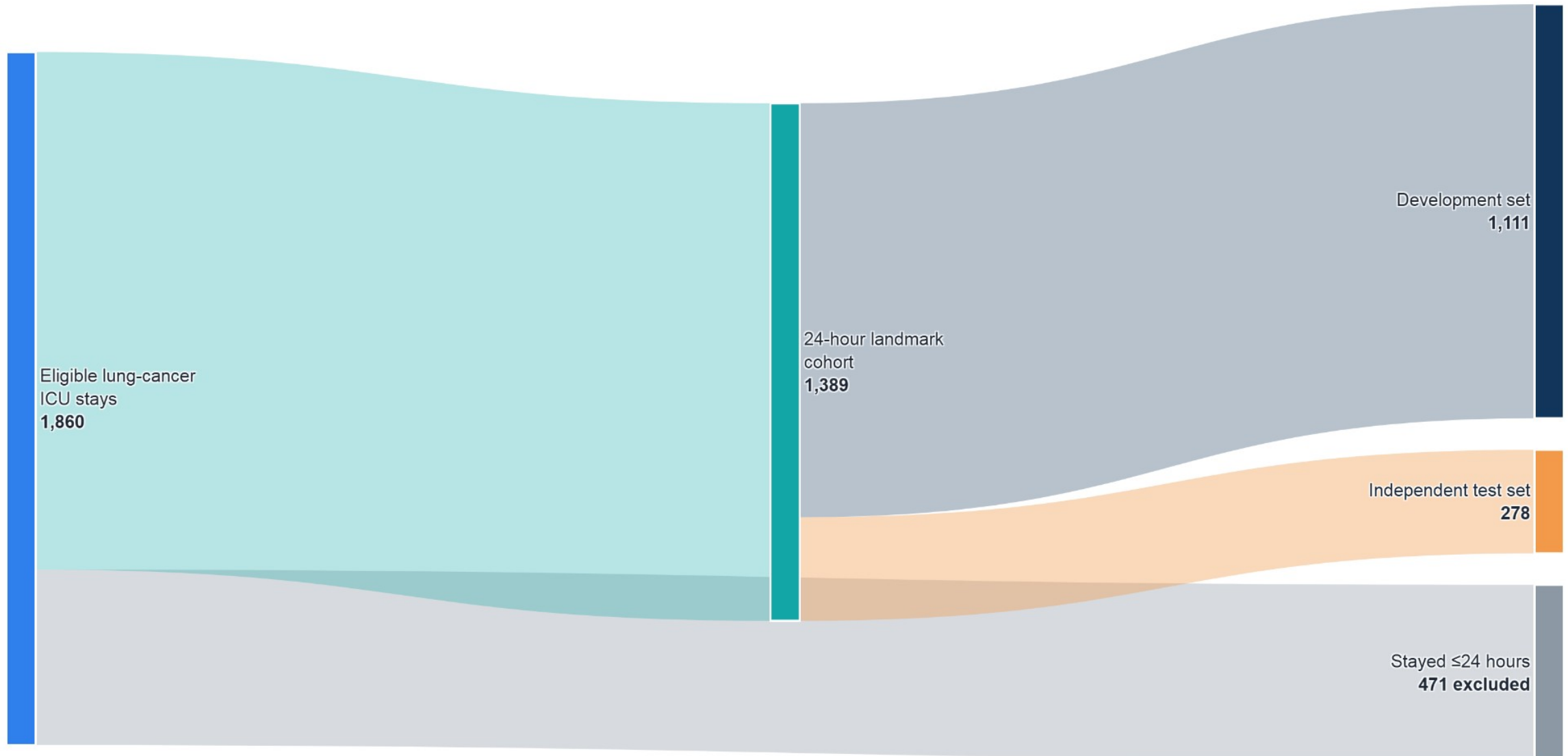
The prediction was made after the first 24 hours

A clear timeline prevents future information from leaking into the prediction



From the source cohort to a fair final test

Patients had to remain in the ICU beyond 24 hours so all predictors were available before prediction



Who were the patients in the study?

Detailed age/sex snapshot of the 24-hour landmark lung-cancer ICU population

Final cohort

1,389



ICU stays after the 24-hour landmark

Outcome split



- Prolonged >5 days
286 (20.6%)
- Short stay ≤5 days
1,103 (79.4%)

Age profile

68 years
median age

IQR: 60-75 years

Older-adult predominant cohort

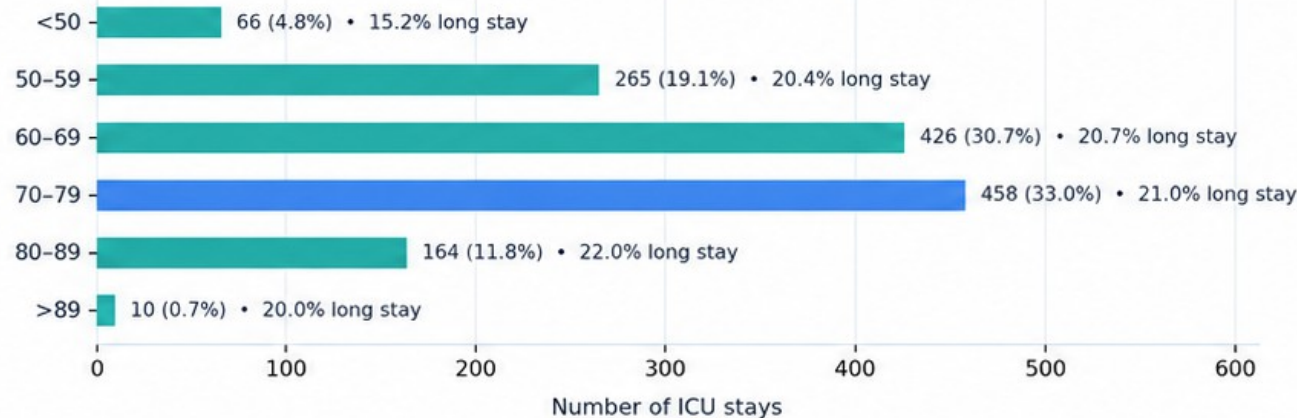
Sex distribution



Female: 661 (47.6%) Male: 728 (52.4%)

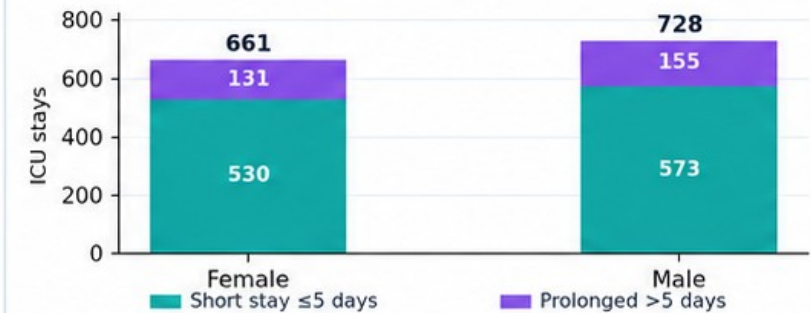
Balanced sex mix; outcome rates were similar

Age distribution



Most patients were 60-79 years old; long-stay rates increased only slightly across age groups.

Sex split and outcomes



Prolonged-stay rate

Female: 19.8%

Male: 21.3%

What did the model see during the first day?

Six clinically meaningful information blocks — no post-24-hour data



Patient & admission

Age, sex, admission source, unit type



Vital signs

Heart rate, oxygen saturation, breathing, blood pressure



Laboratory tests

Creatinine, glucose, sodium, blood counts and more



Medical history

COPD, diabetes, heart disease, cancer therapy



APACHE physiology

Severity indicators from the first ICU day



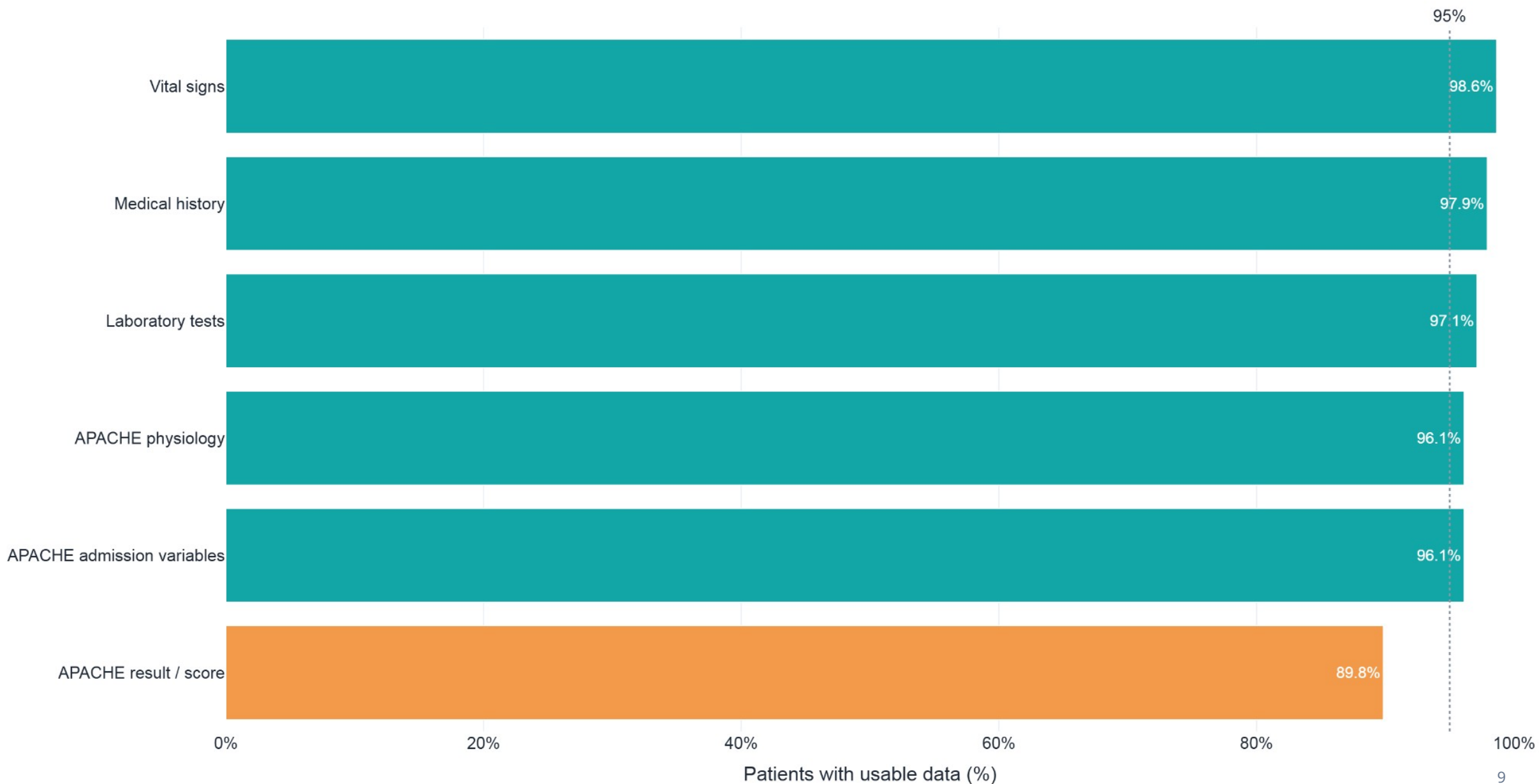
APACHE context

Diagnosis, admission context and severity score

Final Model B feature set: 152 predictors — 141 numeric and 11 categorical

Almost every patient had usable first-day information

High coverage reduces the number of patients lost because of missing data



How did we test the model fairly?

Learn on one group, tune without peeking, then open one untouched test set

10



Learn patterns

1,111

development stays

10



Repeat fairly

5-fold

cross-validation

10



Test once

278

unseen stays

The test set stayed sealed until the model and probability cut-off were locked.

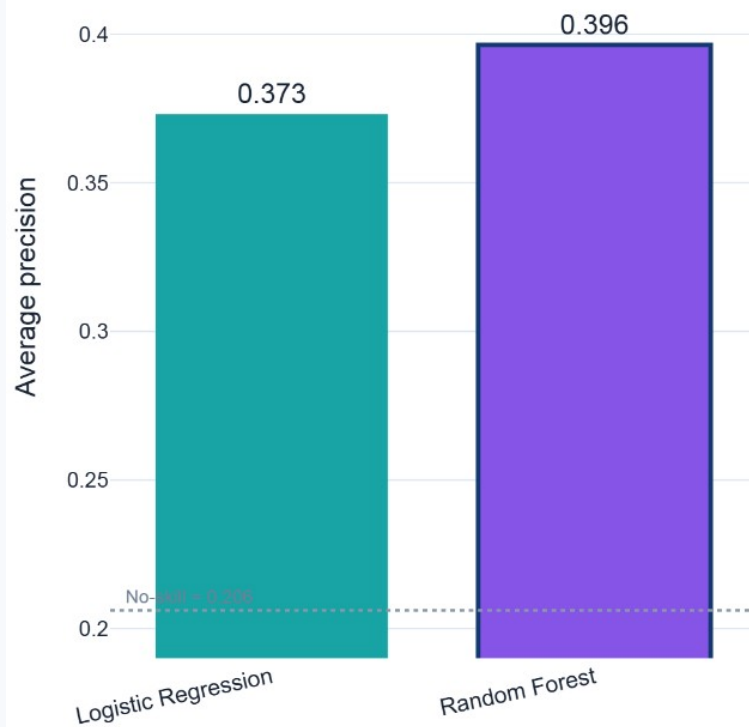
Before tuning, Random Forest looked better

Baseline five-fold cross-validation using clinical + APACHE information

Logistic Regression Random Forest

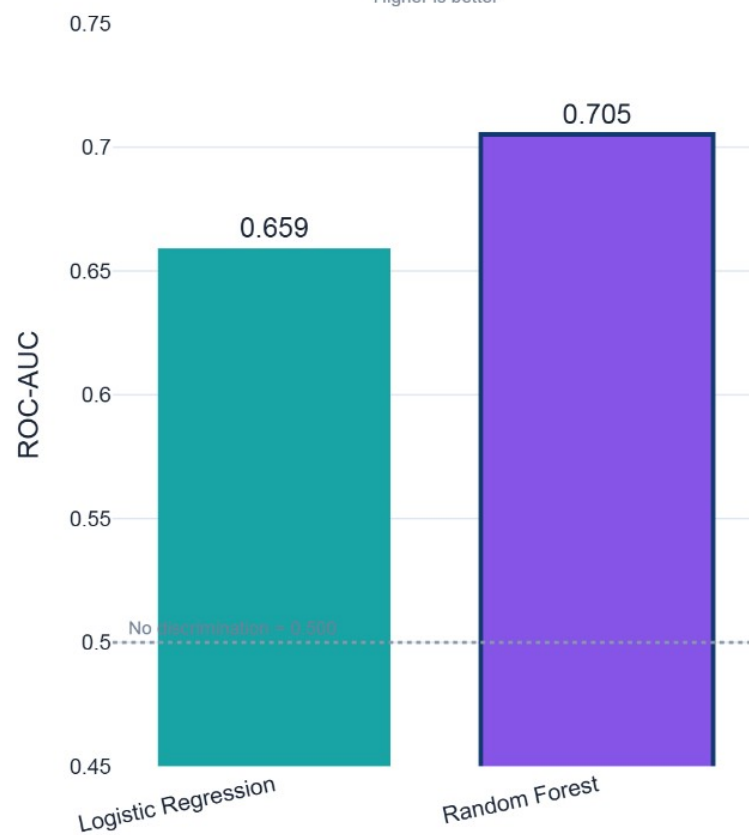
Finding long-stay patients

Higher is better



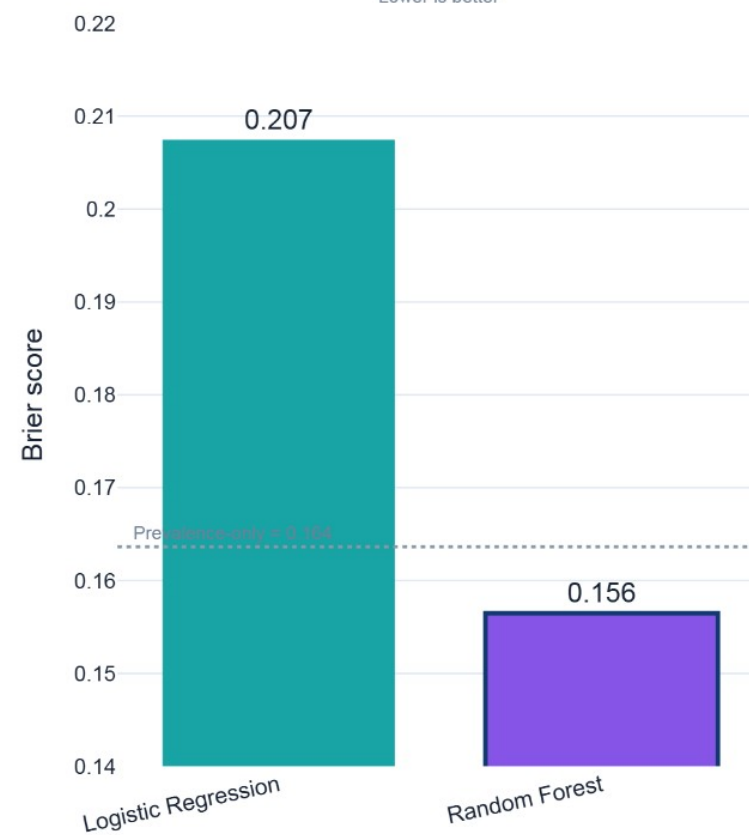
Separating higher from lower risk

Higher is better



Quality of predicted probabilities

Lower is better



Random Forest started ahead in ranking and probability quality.

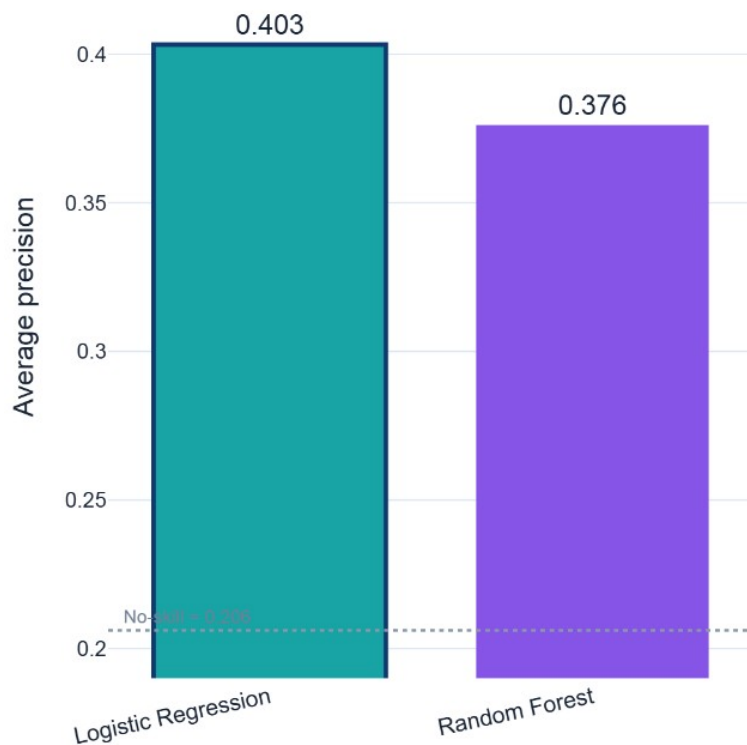
After tuning and calibration, Logistic Regression offered the better overall trade-off

Development out-of-fold results after hyperparameter tuning and sigmoid calibration

Logistic Regression Random Forest

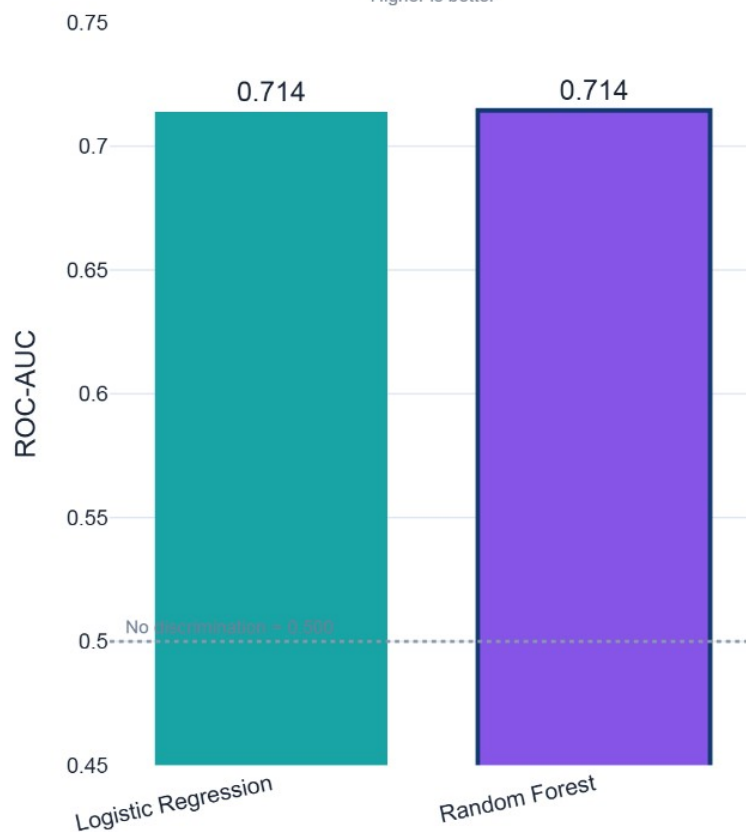
Finding long-stay patients

Higher is better



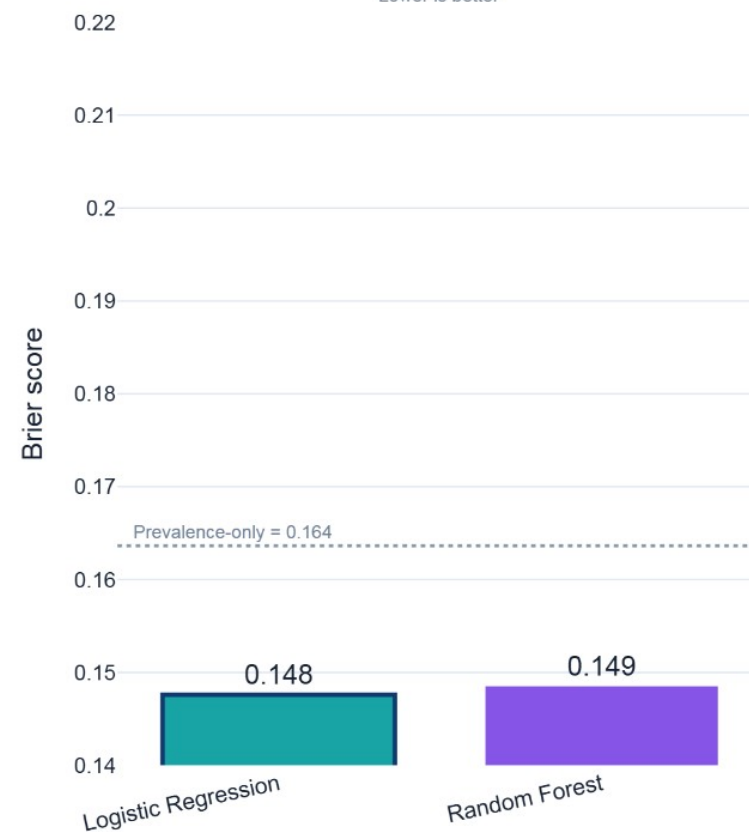
Separating higher from lower risk

Higher is better



Quality of predicted probabilities

Lower is better



Logistic Regression had higher average precision, virtually identical ROC-AUC, a slightly lower Brier score, and much greater interpretability.

The model provided a useful — but not perfect — early warning

Independent test set: 278 patients • 57 stayed in the ICU for more than 5 days

ROC-AUC

0.724

Overall ability to rank higher-risk patients

Average precision

0.412

Quality of the high-risk ranking

Brier score

0.146

Probability error — lower is better

Long stays detected

63.2%

Sensitivity

Short stays recognized

69.2%

Specificity

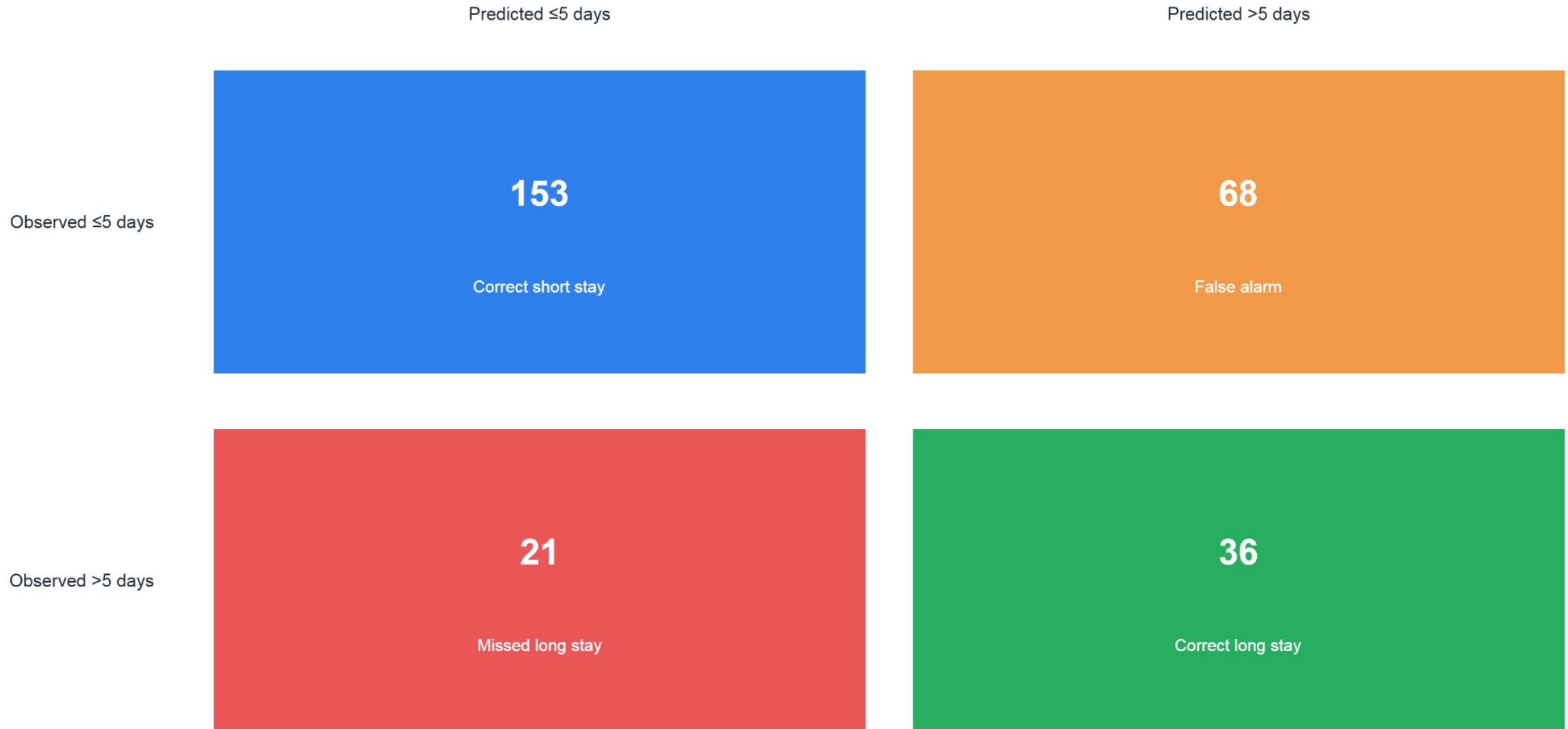
Low-risk reliability

87.9%

Negative predictive value

The model caught 36 of 57 long stays

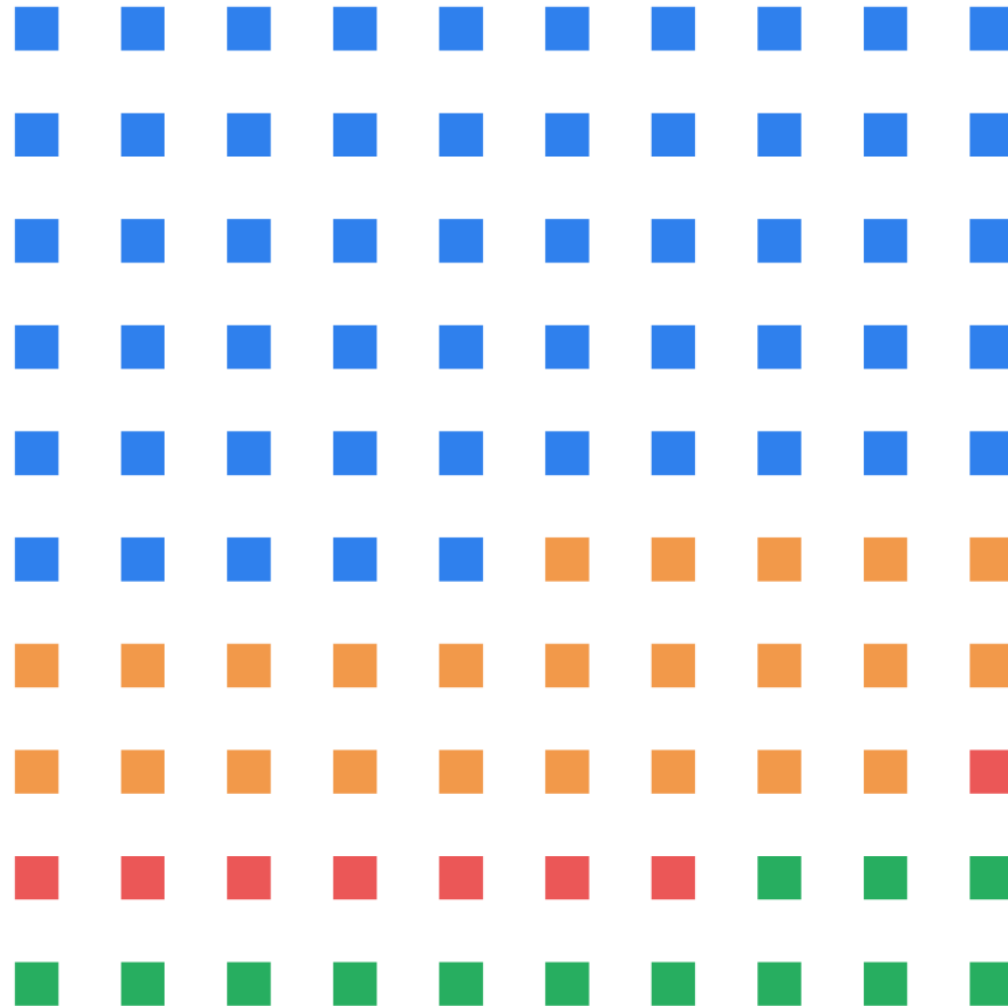
Locked probability cut-off: 0.195 • The cut-off was selected before opening the test set



What the result means for 100 similar patients

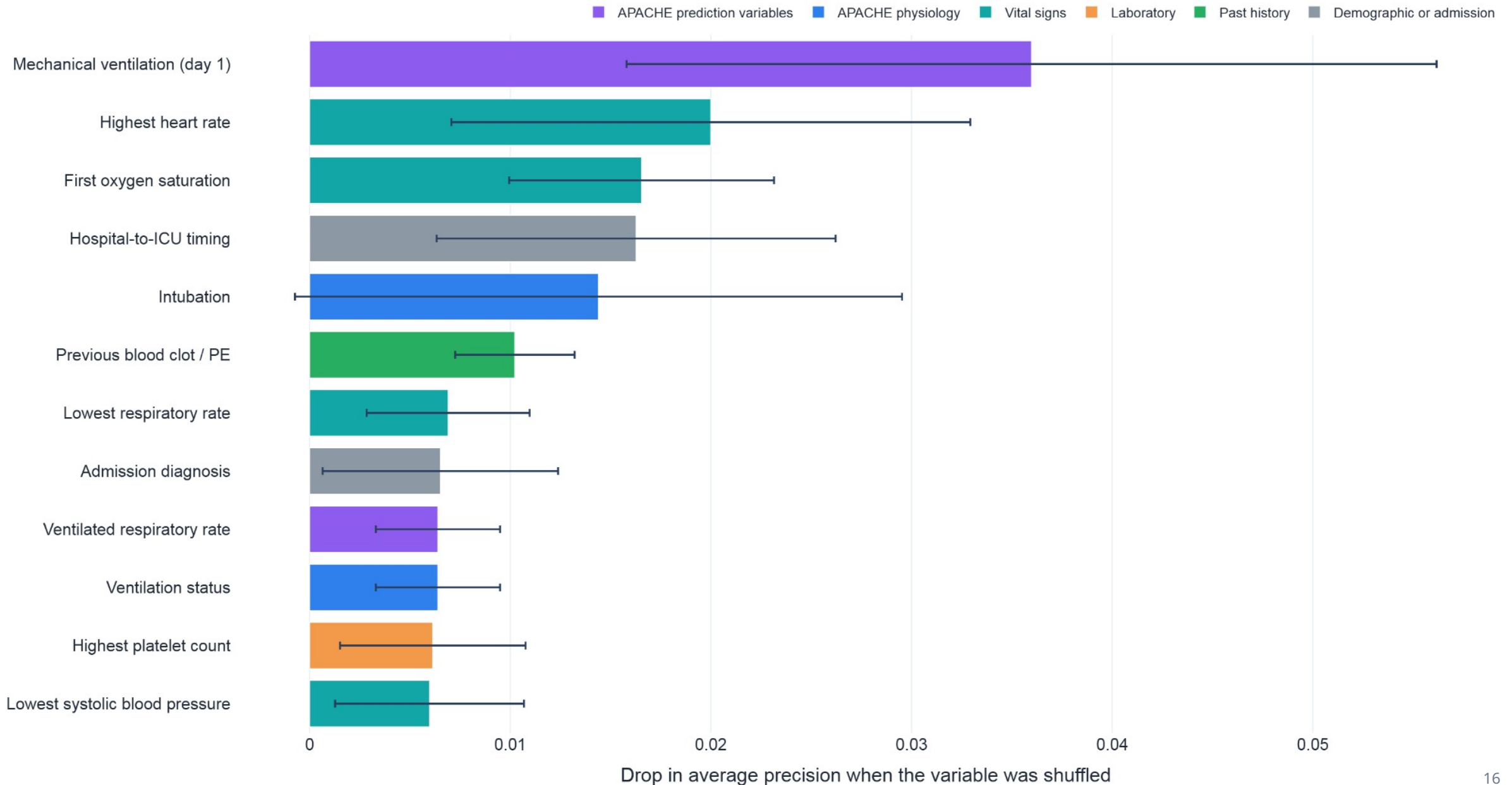
Approximate counts based on the independent test-set performance

■ Correct short stay (55) ■ False alarm (24) ■ Missed long stay (8) ■ Correct long stay (13)



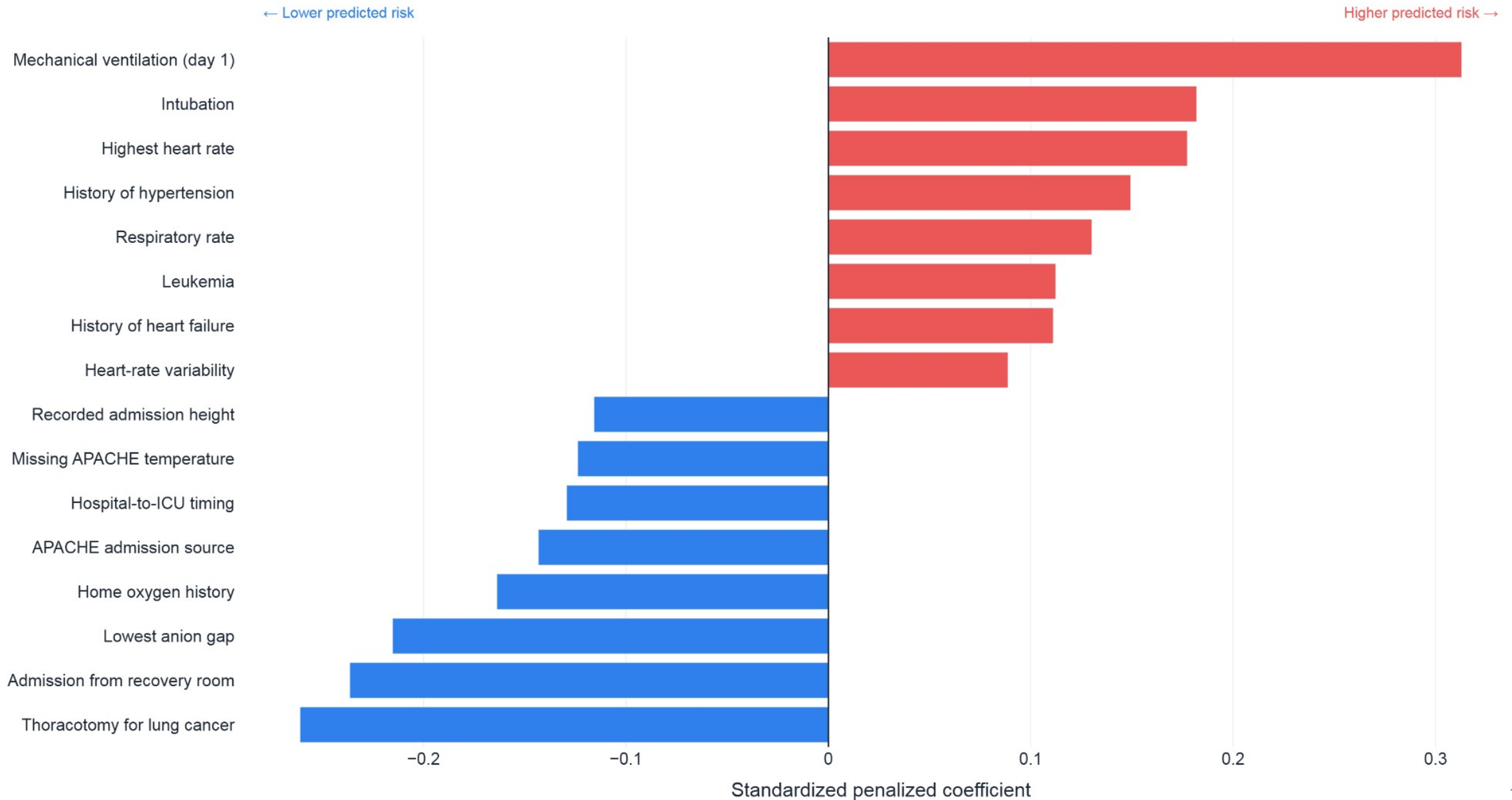
Breathing support and early instability mattered most

Permutation importance measures how much performance falls when one variable is randomly shuffled



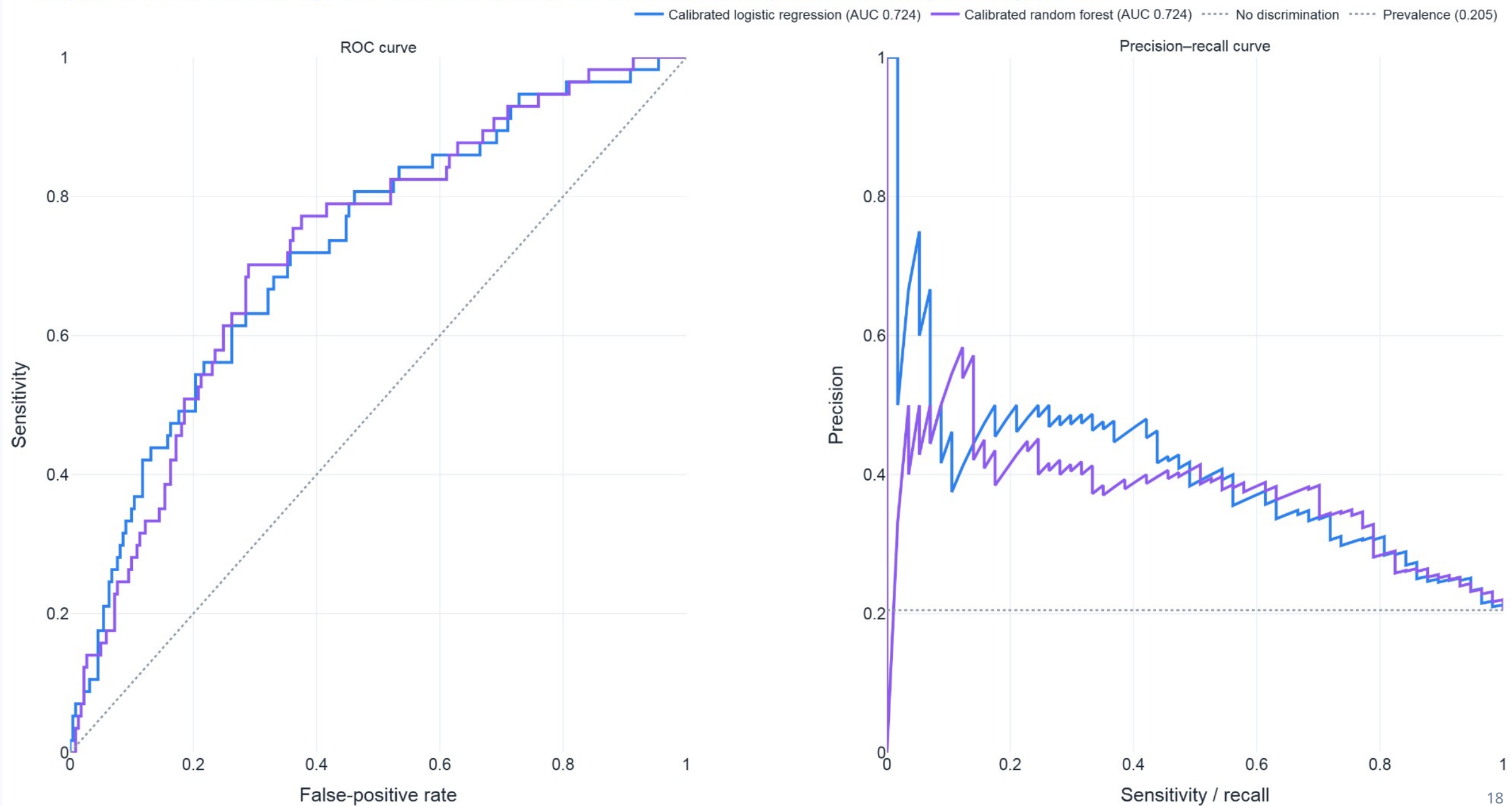
The final model kept only a small set of useful signals

Predictive associations only — they do not prove cause and effect



Both final models separated risk moderately well

Independent test-set curves; the logistic model remained primary because it was more interpretable and had higher average precision



What should we not conclude?

A promising internal result is not the same as a deployable clinical tool



Retrospective data

The study used archived records, not a prospective clinical trial.



Structured lung-cancer identification

Cancer status came from codes and structured diagnosis text.



No external validation yet

The test set was independent, but it came from the same database.



Trade-off at the cut-off

More sensitivity means more false alarms; the best cut-off depends on use.

Next research step: validate prospectively and in a different hospital system.

Three things to remember

20 **The first 24 hours contain useful predictive information.**

The final model achieved ROC-AUC 0.724 on unseen patients.

20 **Respiratory support and early instability mattered most.**

Mechanical ventilation, heart rate and oxygen saturation were key signals.

20 **Useful early warning — not a standalone decision system.**

It can support risk screening, but external validation is still required.



Can we predict a long ICU stay after 24 hours? Yes — moderately well, and early enough to be useful for screening.



Questions?

Thank you

References & visual credits

Appendix

- Pollard, T. J., Johnson, A. E. W., Raffa, J. D., Celi, L. A., Mark, R. G., & Badawi, O. (2018). The eICU Collaborative Research Database, a freely available multi-center database for critical care research. *Scientific Data*, 5, 180178.
- Pollard, T., Johnson, A., Raffa, J., Celi, L. A., Badawi, O., & Mark, R. (2019). eICU Collaborative Research Database (Version 2.0). *PhysioNet*.
- O'Halloran, H. M., Kwong, K., Veldhoen, R. A., & Maslove, D. M. (2020). Characterizing the patients, hospitals, and data quality of the eICU Collaborative Research Database. *Critical Care Medicine*, 48(12), 1737–1743.
- Nature article metrics and PhysioNet access documentation were accessed on 21 July 2026.
- All project charts were generated from aggregate, deidentified results. No patient-level records appear in this presentation.